

Churn Model Training Report

Trained at : 2026-09-11
Training dates : 2026-07-17, 2026-07-22
Holdout date : 2026-07-27
Features : 57 training (9 leakage, 14 dead excluded)
Training samples : 8000 (370 positive)
Holdout samples : 1000 (58 positive)
Class imbalance : 4.62% positive
scale_pos_weight : 20.6216

Metrics (raw probabilities)

Metric	Value
AUC	1.0
5-fold OOF CV AUC	1.0
Brier	0.0016
LogLoss (test)	0.0414
LogLoss (train)	0.0418
Accuracy (optimal)	1.0
ECE (raw)	0.0407
Optimal threshold	0.9509
F1-optimal threshold	0.9509

Classification & Calibration

Confusion Matrix (optimal threshold)

	Pred 0	Pred 1
Actual 0	942	0
Actual 1	0	58

Precision=1.0 Recall=1.0 F1=1.0

Probability Calibration (out-of-fold)

Item	Value
Method	isotonic
ECE raw	0.0406
ECE Platt	0.0002
ECE isotonic	0.0
ECE after	0.0
Brier after	0.0
LogLoss after	0.0

Feature Importances & Hyperparameters

Top-10 Feature Importances

Feature	Importance
txn_count_30d	0.4047
behav_frequency_score	0.3076
behav_active_days_90d	0.2440
amount_growth_ratio	0.0363
txn_count_90d	0.0033
behav_txn_count_7d	0.0030
avg_days_between_txn	0.0007
txn_count_180d	0.0004
total_amount_180d	0.0000
fin_income_growth	0.0000

Hyperparameters

Parameter	Value
max_depth	4
learning_rate	0.05
n_estimators	50
min_child_weight	2
reg_lambda	1.0
subsample	0.8
colsample_bytree	0.8
scale_pos_weight	20.6216

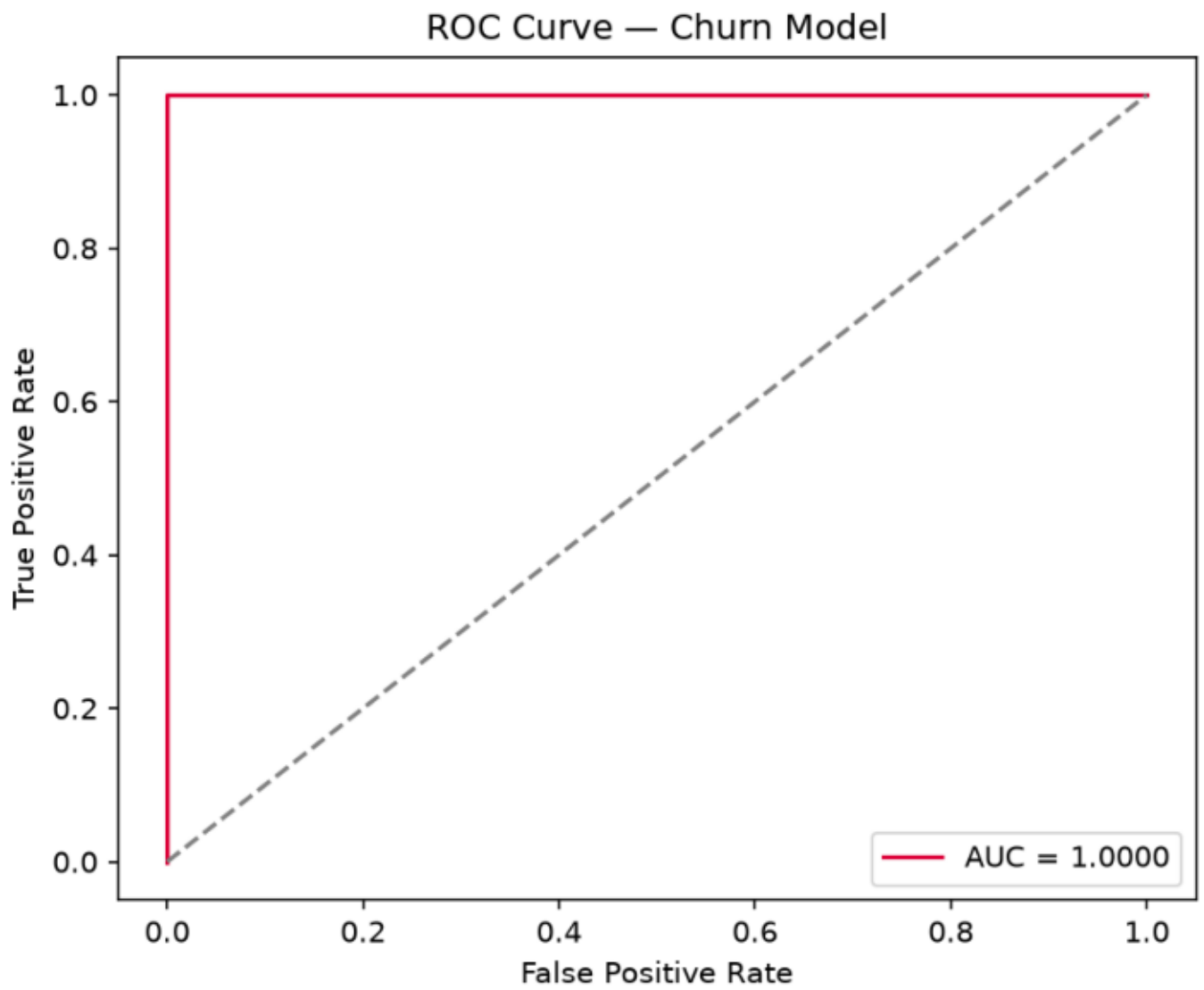
Diagnostics & Drift

train AUC : 1.0
test AUC : 1.0
overfit gap : 0.0 (flag=False)
test missing ratio : 0.0
overconfidence : 0.0

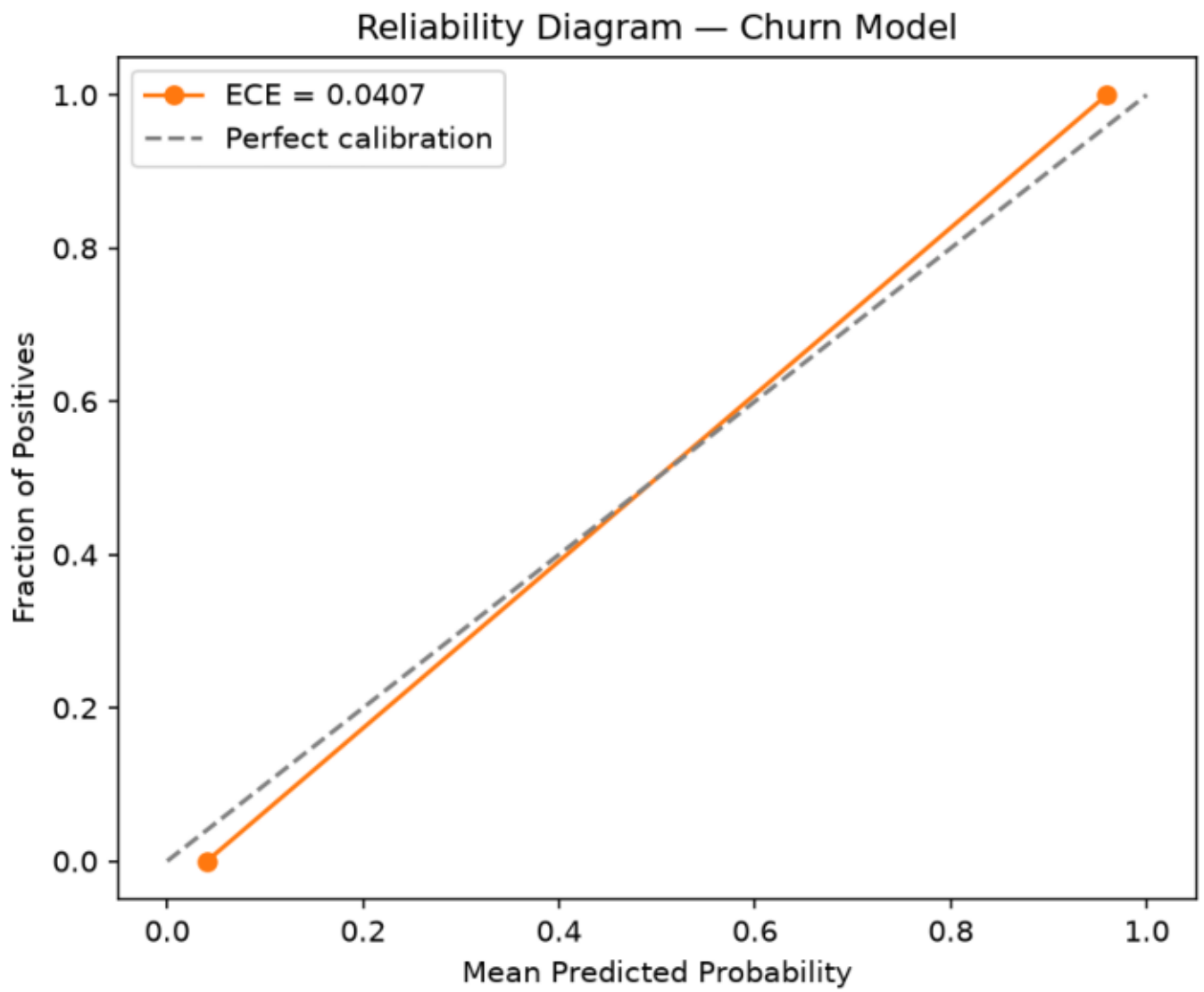
Feature Drift — PSI (max 2.4761, high)

Feature	PSI
days_since_first_txn	2.4761
distinct_channels_90d	0.0888
behav_frequency_score	0.0752
txn_count_30d	0.0752
txn_count_365d	0.0748
txn_count_180d	0.0687
temp_payday_activity_ratio_90d	0.0543
temp_weekday_txn_ratio_90d	0.0506
behav_diversity_score	0.0482
total_amount_180d	0.0447

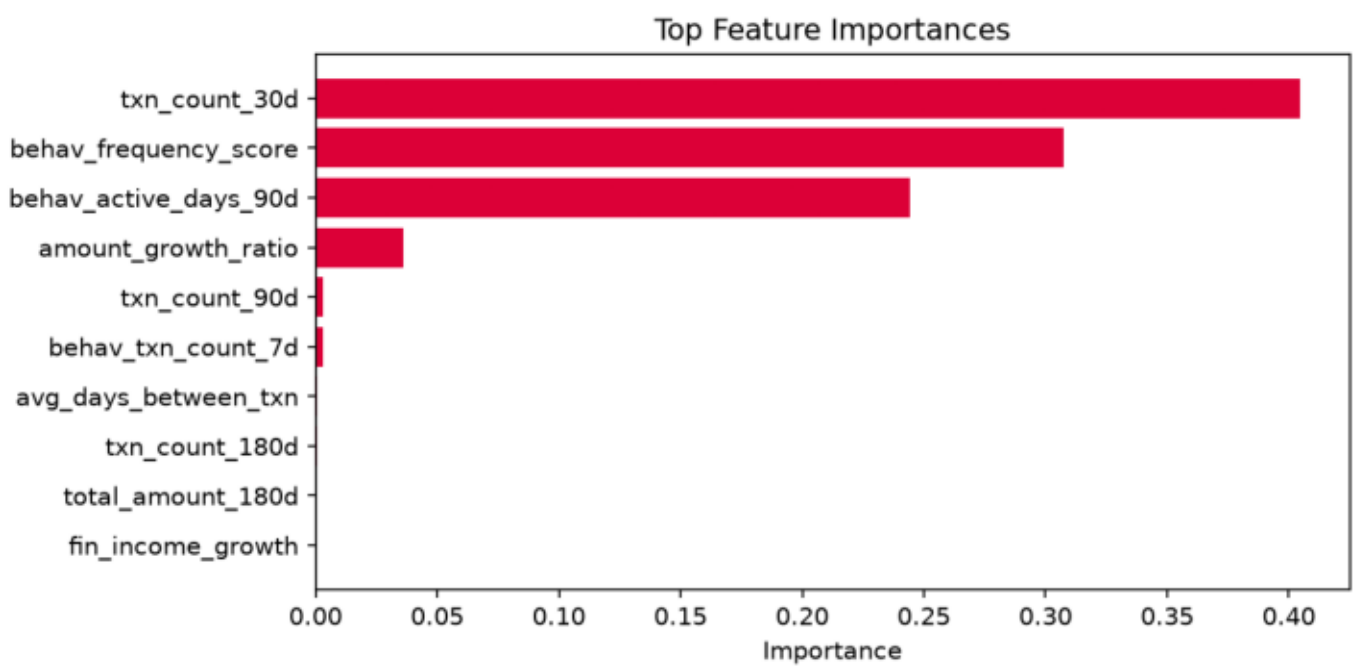
Roc Curve



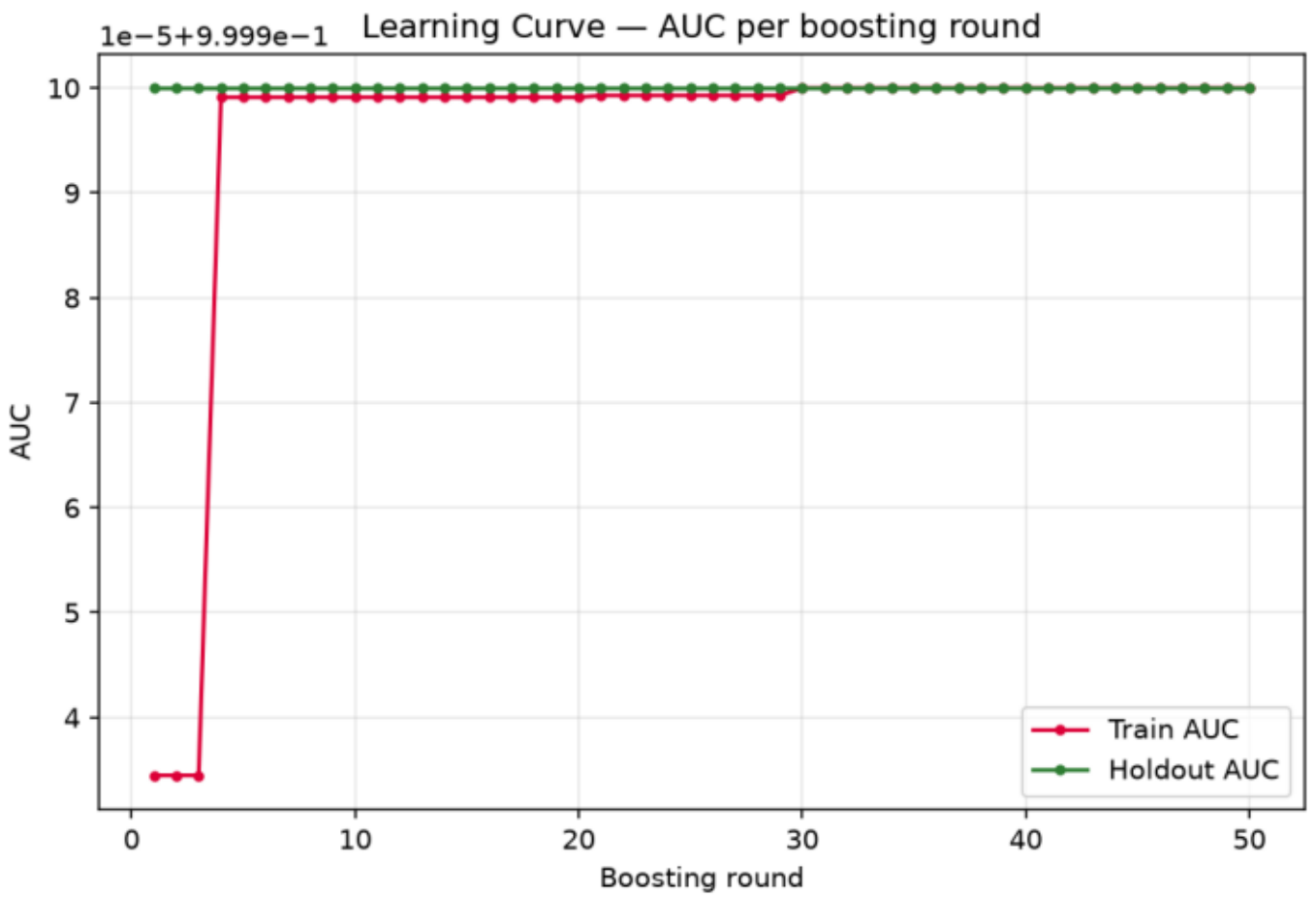
Calibration



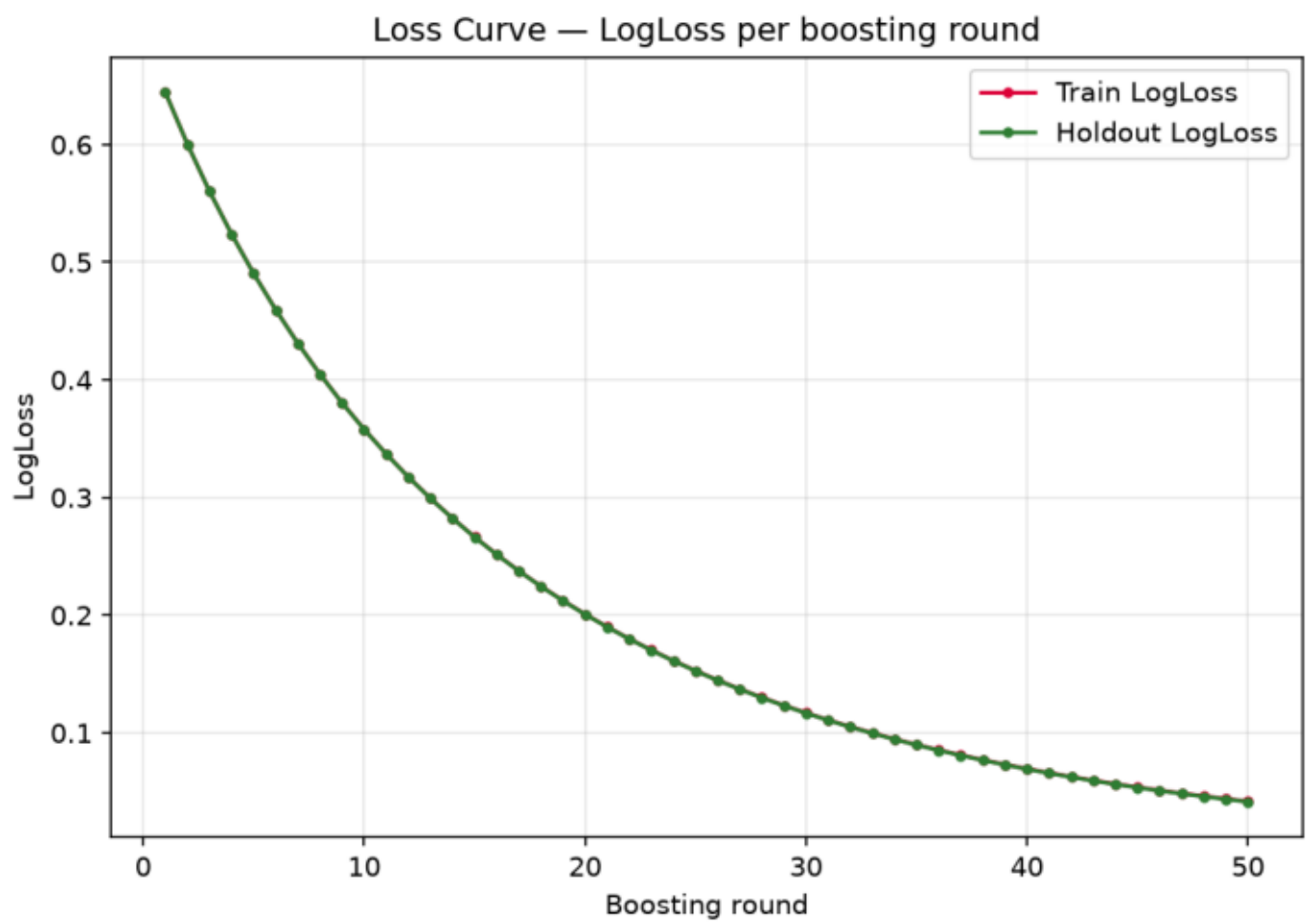
Feature Importance



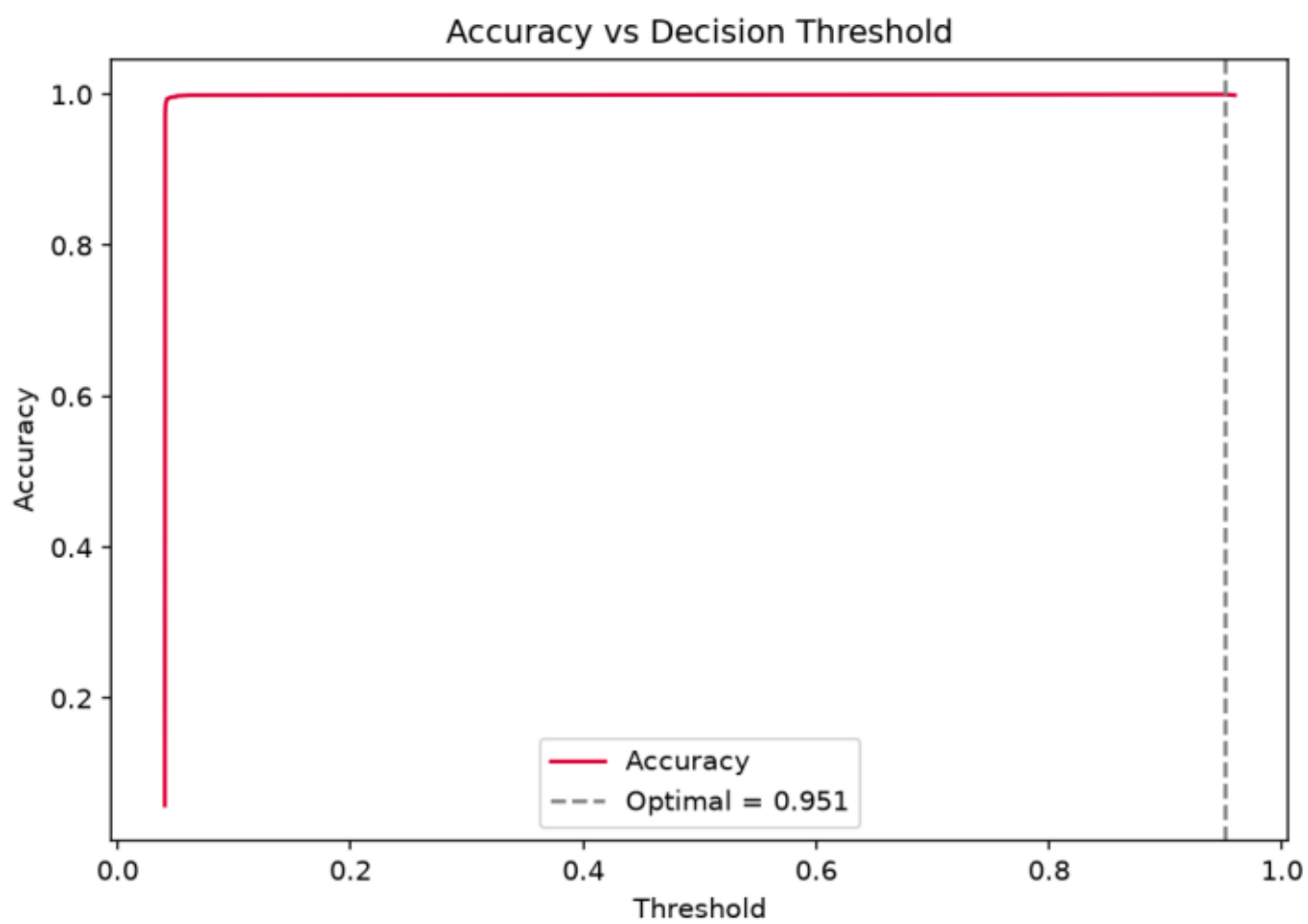
Learning Curve



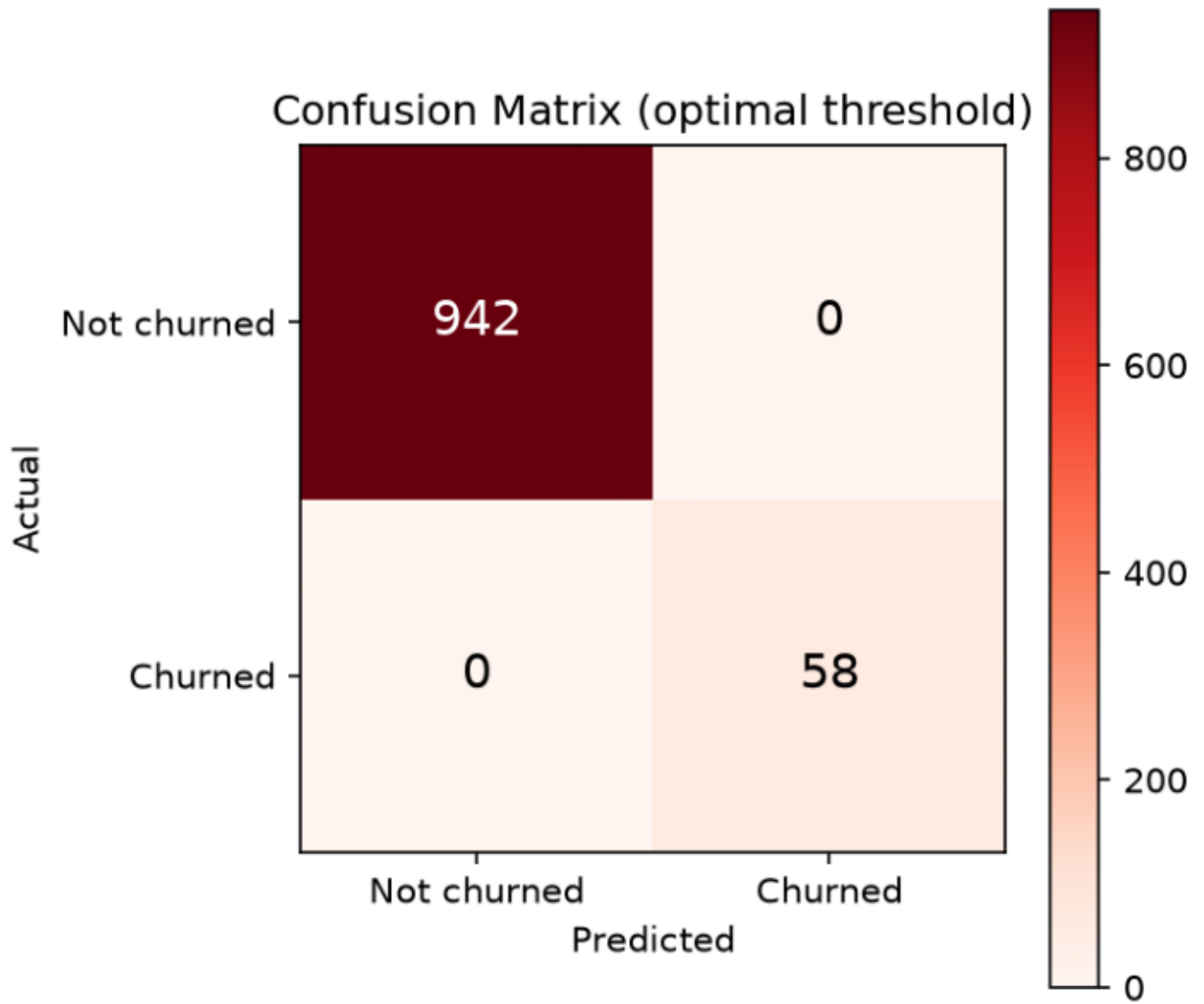
Loss Curve



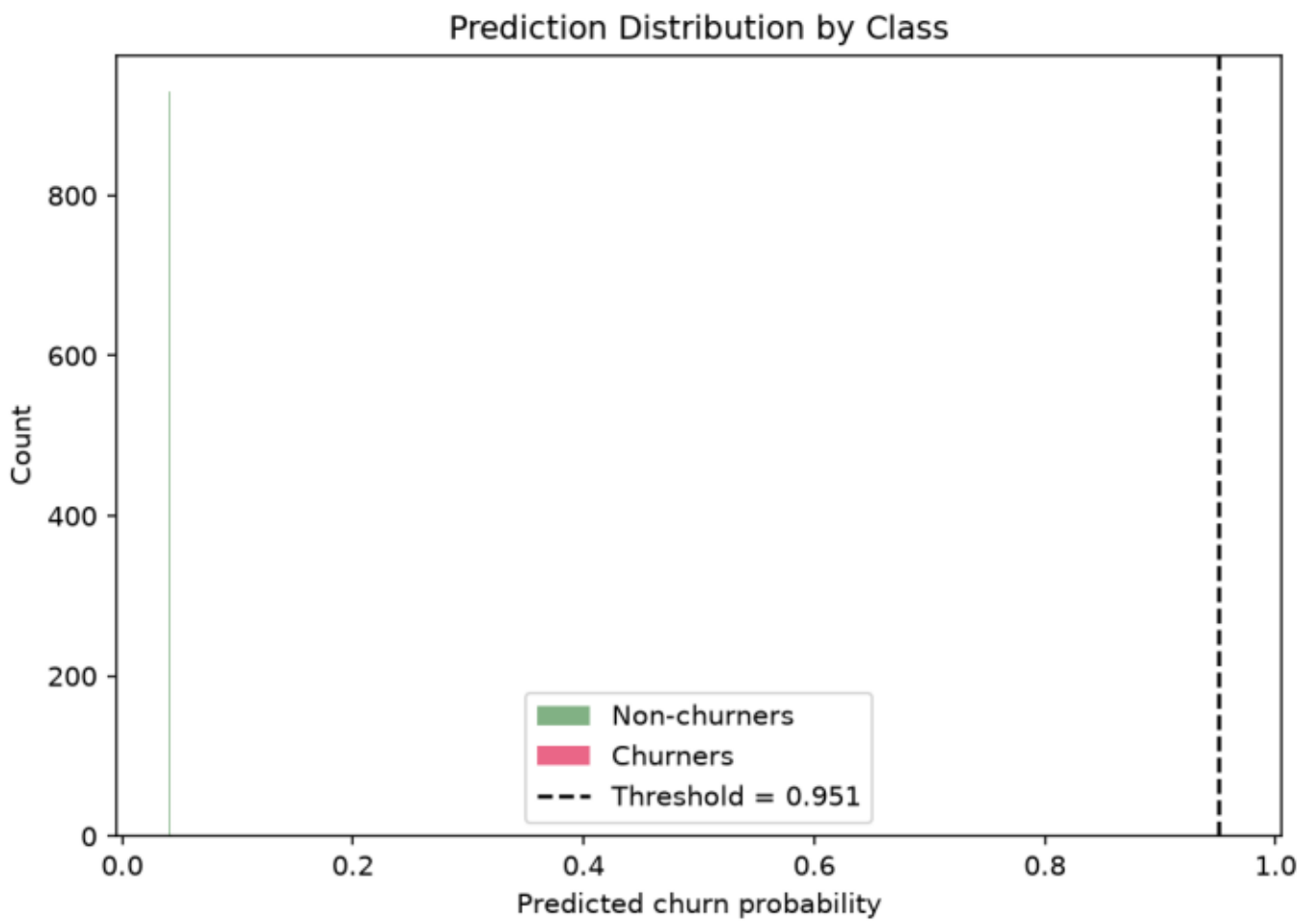
Accuracy Threshold



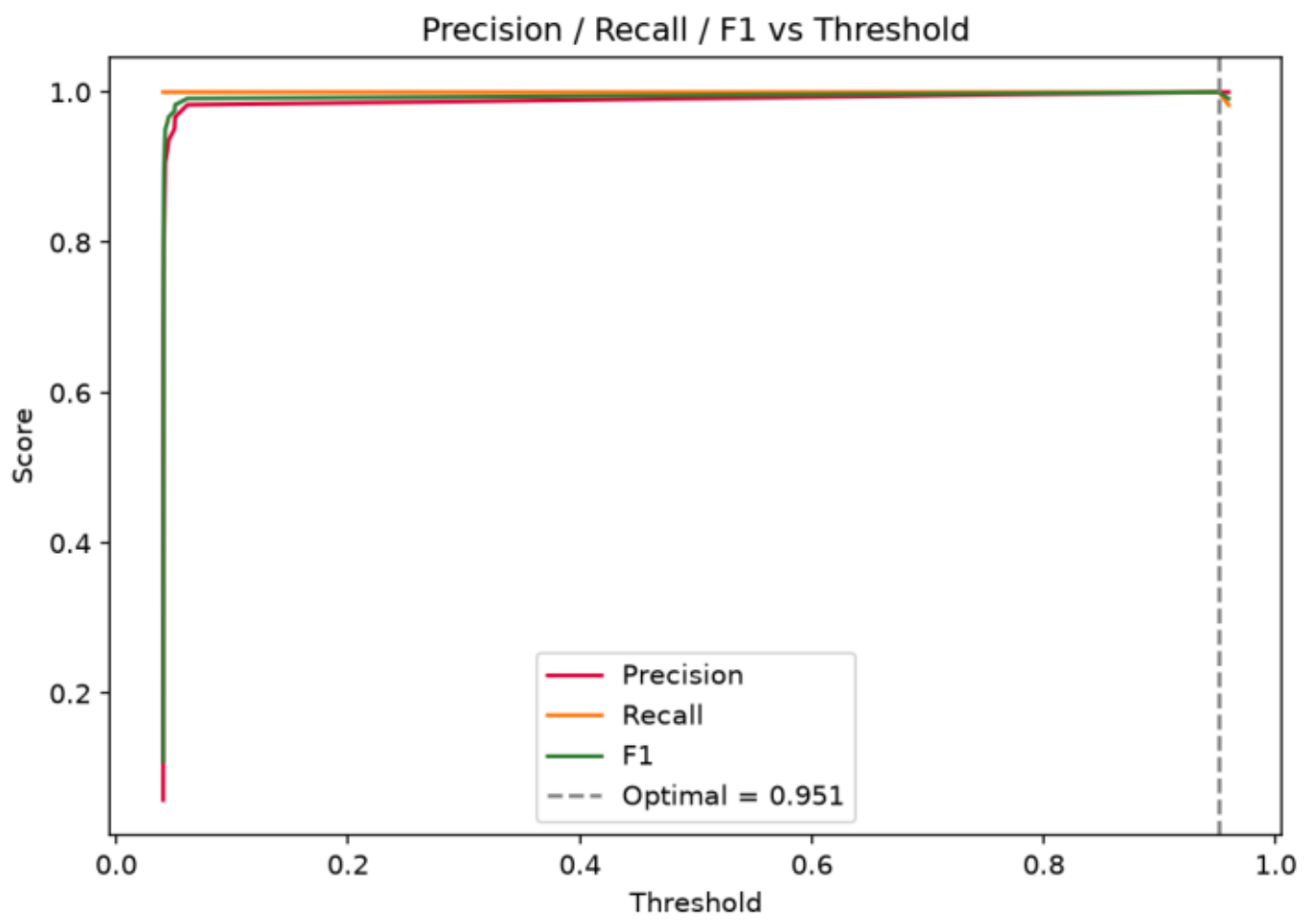
Confusion Matrix



Prediction Distribution



Threshold Analysis



Calibration Comparison

